Deploying MCP server for general access

To enable general access to this MCP server we need to deploy it, i.e., we need to keep the server running somehow so that LLMs like ChatGPT, Claude can access it to get the data we want to provide. There are two modes of this doing this:

1. Self-hosted mode: we keep the code running in our local PC. In this mode, the local PC becomes the server itself and LLMs connect to it directly.
2. Cloud hosting mode: we host our server in the cloud and use the URL as the MCP URL. LLMs hit the cloud to get the data. The cloud hosting will cost a bit. The most cost-efficient one will cost $9 per month.

Self-Hosted Deployment

Let's discuss the setup required for self-hosted mode first.

We need the following tools:

Python

Cloudflared

At first, we navigate to the server code directory. There we create a virtual environment and install all the dependencies. The commands are:

```
python -m venv .venv
.venv\Scripts\activate.bat
pip install -r requirements.txt
```

Then we install Cloudflared CLI. For windows users, the command is:

```
winget install --id Cloudflare.cloudflared
```

the we run the MCP server locally

```
$env:MCP_TRANSPORT="streamable-http"; $env:PORT="8000"
python server.py
```

if the server runs successfully, we tunnel it through cloudflared

```
cloudflared tunnel --url http://localhost:8000
```

Once the tunnel is successful, it will show a URL like: https://something-random.trycloudflare.com/mcp

We will use this URL as the URL of our custom MCP server for Claude, ChatGPT, etc.

Adding the MCP server to Claude

For Claude, we navigate to Settings and then click Connectors tab. There, if we click the add button, it will show “add custom connector.” Then we can insert a desired name of our custom server and the URL provided by cloudflared tunnel. Once click continue, it will ask us about the authentication mechanism. There, we will select, No Sign-In option and click add. This should add our MCP server to claude.

Once it is done, we can open another chat window and ask any question we would like an answer to.

Adding the MCP server to ChatGPT

For ChatGPT, it is kind of similar. We go to Settings and then Plugins. From Plugins, we select Browse Directory. In that page there will be an Add option on the top right corner. Upon clicking Add, we will see three options in a drop-down list. From those options, we will select “Add custom MCP server”. There we will provide a desired name of the server and then the URL from the Cloudflared. This will create our plugin. In order to use it in a chat, one must use @ followed by the name of the server, in order to fetch the data from our MCP server.